

Macaev Liftability Is Equivalent to Logarithmic Local Linearization

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Status. Candidate partial result, likely valid. The universal problem is reduced exactly to a sharp finite-dimensional estimate, but that estimate is not decided here.

1 Source question

Cabello Sanchez and Garcia ask on page 5 of [1] whether every operator in the Macaev ideal is liftable through every twisted Hilbert extension. Their Macaev ideal consists of compact operators T such that

$$\sum_{j \geq 1} \frac{s_j(T)}{j} < \infty.$$

Note that $\|F\|$ may depend not only on the norms of the summands, but also on the Banach-Mazur distance between $\mathcal{H} \oplus_\phi \mathcal{H}$ and its Banach envelope which, in turn, depends only on $Q(\phi)$. □

It is clear that an operator belongs to *some* $\mathcal{S}_p$ if and only if its singular numbers are $O(n^{-\alpha})$ for *some* $\alpha > 1$, which does not necessarily agree with $1/p$. An ideal that contains all the Schatten classes and is commonly regarded as reasonably small is the Macaev ideal $\mathcal{S}_\omega$ of those operators whose singular values satisfy $\sum_n s_n/n < \infty$, with the obvious norm, see [15, 8]. This ideal appears naturally in a surprisingly wide variety of situations, problems, results and applications, as a quick internet search reveals.

QUESTION. Is every operator in the Macaev ideal liftable?

Figure 1: The source question in arXiv:2112.03852, page 5.

For a homogeneous quasilinear map $\Phi : E \rightarrow F$, let $Q(\Phi)$ be the least constant in

$$\|\Phi(x + y) - \Phi(x) - \Phi(y)\| \leq Q(\Phi)(\|x\| + \|y\|),$$

and put

$$\text{dist}(\Phi, \text{Lin}) = \inf_{L \text{ linear}} \sup_{x \neq 0} \frac{\|\Phi(x) - Lx\|}{\|x\|}.$$

Fixing a separable infinite-dimensional Hilbert space $\mathcal{H}$, define

$$\Lambda(m) = \sup\{\text{dist}(\Phi, \text{Lin}) : \Phi : \ell_2^m \rightarrow \mathcal{H}, Q(\Phi) \leq 1\}.$$

This number is finite. Indeed, subtract the linear map agreeing with Φ on an orthonormal basis and combine the coordinates in a balanced binary tree. At each level Cauchy-Schwarz bounds the sum of the norms of the partial coordinate blocks. Summing the resulting geometric series gives, for example, $\Lambda(m) \leq 4\sqrt{m}$.

2 Idea of the proof

The positive direction is Abel summation. A bound $\Lambda(m) = O(\log(m+1))$ gives logarithmically bounded liftings of rank- m projections. The spectral decomposition

$$D = \sum_{m \geq 1} (s_m - s_{m+1}) P_m$$

then becomes a convergent series of liftings precisely because the Macaev condition is the discrete integral of those logarithmic costs.

The converse uses the same mechanism in reverse. If the logarithmic bound fails, select increasingly bad finite-dimensional quasilinear maps and place them on orthogonal blocks. Scalar coefficients are chosen so that their Macaev contributions are summable while their block linearization obstructions tend to infinity. The resulting block-scalar Macaev operator cannot lift through the completed block twist.

3 The endpoint dichotomy

Theorem 1. *The following are equivalent.*

1. *Every operator in the Macaev ideal on a separable Hilbert space is liftable through every twisted Hilbert extension.*
2. *There is a universal constant C such that*

$$\Lambda(m) \leq C \log(m+1) \quad (m \geq 1).$$

More precisely, if (2) fails, a block direct-sum twist and a block-scalar Macaev operator give a counterexample to (1).

Proof. Assume (2). By the ideal property of liftable operators [1, Lemma 1], it is enough to treat a positive diagonal compact operator D with decreasing diagonal (s_m) . Let P_m be the projection onto its first m eigenvectors and set $d_m = s_m - s_{m+1}$. Thus

$$D = \sum_{m \geq 1} d_m P_m.$$

Fix a twisted Hilbert extension and represent it by a quasilinear map $\Phi : \mathcal{H} \rightarrow \mathcal{H}$. Restricting Φ to $P_m \mathcal{H}$ and using (2), choose a linear map $L_m : P_m \mathcal{H} \rightarrow \mathcal{H}$ such that

$$\|\Phi(P_m x) - L_m P_m x\| \leq CQ(\Phi) \log(m+1) \|x\|.$$

Hence $x \mapsto (L_m P_m x, P_m x)$ is a lifting of P_m whose norm, in the standard twisted-sum quasinorm, is at most $1 + CQ(\Phi) \log(m+1)$.

Summation by parts gives

$$\begin{aligned} \sum_{m \geq 1} d_m \log(m+1) &= s_1 \log 2 + \sum_{m \geq 2} s_m (\log(m+1) - \log m) \\ &\leq s_1 \log 2 + \sum_{m \geq 2} \frac{s_m}{m} < \infty. \end{aligned}$$

A twisted Hilbert space has an equivalent Banach norm. Therefore the weighted series of the displayed liftings converges to a bounded linear lifting of D . This proves (1).

We prove the converse contrapositively. Suppose that $\Lambda(m)/\log(m+1)$ is unbounded. Choose strictly increasing integers m_k and maps

$$\Phi_k : \ell_2^{m_k} \rightarrow Y_k, \quad Q(\Phi_k) \leq 1,$$

where the Y_k are Hilbert spaces, so that

$$\delta_k := \text{dist}(\Phi_k, \text{Lin}) \geq k^4 \log(m_k + 1).$$

On the algebraic Hilbertian direct sum define

$$\Omega((x_k)) = (\Phi_k(x_k)).$$

Minkowski's inequality shows that $Q(\Omega) \leq 1$. Completing the associated twisted sum gives an exact sequence whose quotient is $X = (\bigoplus_k \ell_2^{m_k})_2$ and whose kernel is $Y = (\bigoplus_k Y_k)_2$.

Set

$$a_k = \frac{1}{k^2 \log(m_k + 1)}, \quad D = \bigoplus_{k \geq 1} a_k I_{\ell_2^{m_k}}.$$

The numbers a_k decrease to zero, so D is compact. Put $N_k = \sum_{i \leq k} m_i$. Since the singular value a_k occurs m_k times, the contribution of the k -th block, for $k \geq 2$, satisfies

$$\sum_{j=N_{k-1}+1}^{N_k} \frac{s_j(D)}{j} = a_k \sum_{j=N_{k-1}+1}^{N_k} \frac{1}{j} \leq a_k \log \frac{N_k}{N_{k-1}} \leq a_k \log(m_k + 1) = \frac{1}{k^2}.$$

The first block contributes only a finite amount, and therefore D belongs to the Macaev ideal.

If D had a bounded lifting through the completed block twist, restriction to the k -th quotient block and projection onto the k -th kernel block would give a linear $L_k : \ell_2^{m_k} \rightarrow Y_k$ with

$$\sup_k \|a_k \Phi_k - L_k\| < \infty.$$

By homogeneity this forces $\sup_k a_k \delta_k < \infty$. But

$$a_k \delta_k \geq k^2,$$

a contradiction. Thus D is a Macaev operator which is not liftable, so (1) fails. $\square$

4 A sharp finite-range theorem

The full logarithmic estimate remains undecided when the target Hilbert space is unrestricted. There is, however, a sharp theorem when both dimensions are controlled. Let

$$\Lambda_c(m, r) = \sup \{ \text{dist}(\Phi, \text{Lin}) : \Phi : \ell_2^m \rightarrow \ell_2^r \text{ continuous and homogeneous, } Q(\Phi) \leq 1 \}.$$

Theorem 2. *There is a universal constant C such that*

$$\Lambda_c(m, r) \leq C(1 + \log(m + r)).$$

This order is sharp when m and r are comparable. Consequently, fix a continuous quasilinear map $\Phi : \mathcal{H} \rightarrow \mathcal{H}$ and nested finite-dimensional subspaces $E_m \subset \mathcal{H}$, $\dim E_m = m$. If, for some fixed $a < \infty$,

$$\dim \text{span } \Phi(E_m) \leq (m+1)^a \quad (m \geq 1),$$

then every positive diagonal Macaev operator whose first m eigenvectors span E_m lifts through the twist generated by Φ .

Proof. Put $n = \max(m, r)$, embed domain and range isometrically in ℓ_2^n , and let $P : \ell_2^n \rightarrow \ell_2^m$ be the orthogonal projection. The map

$$\tilde{\Phi}(x) = J\Phi(Px) : \ell_2^n \longrightarrow \ell_2^m$$

is continuous, homogeneous, and has quasilinearity constant at most one. Kalton's logarithmic linearization theorem [2, Theorem 2.2] supplies a linear $A : \ell_2^n \rightarrow \ell_2^m$ such that

$$\|\tilde{\Phi}(x) - Ax\| \leq C(1 + \log n)\|x\|.$$

Restricting A to the embedded ℓ_2^m and orthogonally projecting its values to the embedded ℓ_2^r proves the upper bound. Kalton's example [2, Proposition 2.1] gives the matching logarithmic lower order when the dimensions are comparable.

For the consequence, restrict Φ to E_m and identify the Hilbert space $\text{span } \Phi(E_m)$ with $\ell_2^{r_m}$, where $r_m \leq (m+1)^a$. The first part yields linear maps $L_m : E_m \rightarrow \mathcal{H}$ satisfying

$$\|\Phi(x) - L_mx\| \leq C_a Q(\Phi) \log(m+1)\|x\| \quad (x \in E_m).$$

The Abel-summation proof in Theorem 1 now applies verbatim. □

5 Verification and limitation

The proof uses only two-sided estimates: Abel summation for the positive direction and restriction to individual blocks for the negative direction. The dense block map is used only to define its completed twisted sum, exactly as in standard block constructions of twisted Hilbert spaces. No uniform boundedness of the individual maps is assumed or needed.

The endpoint dichotomy identifies the remaining issue sharply, but does not decide it. Theorem 2 shows that target dimension is the only unresolved quantitative parameter in the continuous finite-dimensional problem: Kalton's theorem gives $O(\log(m+r))$, whereas the universal Macaev question requires $O(\log m)$ with no restriction on r , including an infinite-dimensional target. A dualization followed by Maurey's projection theorem does not remove this parameter: estimating the required type constant is quantitatively equivalent to bounding the original splitting constant. The available Schatten-class argument in [1] supplies a bound depending on every fixed exponent $p < \infty$; its cited splitting theorem is nonquantitative in p . Consequently it does not currently yield the linear growth in p that would imply (2). Human review should concentrate on the completed block construction and on whether an existing local theory theorem already decides the asymptotic behavior of $\Lambda(m)$.

References

- [1] F. Cabello Sanchez and R. Garcia, *The Twisted Hilbert Space Ideals*, Integral Equations Operator Theory 94 (2022), article 21; arXiv:2112.03852.
- [2] N. J. Kalton, *A remark on quasi-isometries*, Proc. Amer. Math. Soc. 131 (2003), 1225–1231.